Stefano Baldan

Staff Software Engineer · Frontend Platform & Accessibility

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https://stefanobaldan.com · https://github.com/singintime

PROFILE

Staff software engineer with over ten years in frontend, specialising in platform architecture for large-scale web applications. Led accessibility engineering for Miro's HTML Canvas–based product for five years, steering it toward WCAG 2.2 AA and European Accessibility Act compliance and retaining eight-figure ARR. Designed the AI-ready standards format that lets domain owners scale quality across a 400+ contributor monorepo. Background in digital signal processing, music, sound synthesis and multimedia computing.

WORK EXPERIENCE

Staff Software Engineer · Miro — Apr 2024 – present

- Resolved accessibility blockers escalated directly by enterprise customers at risk of churning, retaining contracts worth eight-figure ARR.
- Led AI-driven developer experience for frontend quality standards across a monorepo with 400+ active contributors: designed a machine-readable format for package instructions and specs that domain owners extend for accessibility, localization, design system and content design, so AI coding agents produce compliant code by default.
- Delivered voice dictation end to end: core service, React components, AI chat integration, session token refresh, and screen-reader-verified error handling.
- Mentored a mid-level engineer through promotion to Senior and a Senior engineer on the path to Staff; conducted ~100 live-coding and system-design interviews and trained new interviewers.
- Drove self-initiated, AI-assisted tech-debt removal across team boundaries: removed jQuery from the entire web client (36 PRs through code owned by 26 teams, −13.8K lines), and migrated dialogs across 14 teams' code to the shared accessible package, retiring the deprecated dialog service and its recurring focus-management bugs.
- Created 12 shared packages, including the canvas accessibility layer (adopted by 14 packages), its keyboard-support framework for interactive canvas objects (adopted by 8 packages, from structured documents to AI workflow nodes) and reduced-motion support (13 packages).
- Fixed screen-reader-triggered slowdowns on very large boards: 3x faster bounds computation for 10K-object selections (45 ms to 15 ms), plus bounds caching and debounced announcements.

Frontend Software Engineer · Miro — Nov 2021 – Mar 2024

- Made Miro's HTML Canvas, invisible to standard web accessibility technologies, operable by keyboard, screen reader and voice-control users, driving the work from research and design through implementation.
- Designed and built keyboard navigation across board objects: spatial navigation for free-form, non-grid layouts (nearest object in four directions) and a semantic tab order.
- Developed a DOM overlay that exposes Canvas content to voice-control software, so users can act on board objects by voice.
- Built alternative views of board text and images, ordered by their semantic position on the board.
- Improved rendering performance in the Miro client using the Canvas and WebGL APIs.

Engineering Team Lead · byrd — Jan 2021 – Oct 2021

Logistics company - systems for shipping parcels from origin to destination

- Led and managed a full-stack team of 4 engineers.
- Frontend development in Angular and TypeScript.
- Backend microservices built with Python, Flask, and SQLAlchemy.

Senior Frontend Engineer · byrd — Nov 2019 – Dec 2020

- Developed a single-page application for the company's Warehouse Management System.
- Built reactive pages and UI components in Angular and TypeScript.

Frontend Developer · Contralto Audio — Sep 2017 – Dec 2019

- Developed a 3D editor for the acoustic simulation of loudspeaker positioning in live venues.
- Created custom 3D graphics components and GPU shaders on top of Three.js and WebGL, shipped as a desktop application with Angular and Electron.

Research Assistant · Iuav University of Venice — Jan 2014 – Dec 2017

- Research for the EU project SkAT-VG (Sketching Audio Technologies using Vocalizations and Gestures): https://skatvg.unipa.it/
- Leader of the Work Package on imitation-driven sound synthesis.
- Developed physically-informed sound synthesis and analysis algorithms in Python and C/C++.

Freelance IT Consultant · Self-employed — Oct 2011 – Dec 2013

- Web-based edutainment games for Pearson Italia; interactive media and sound design for digital art installations.

TECHNICAL SKILLS

Frontend: TypeScript, React, JavaScript, HTML Canvas, WebGL, Three.js, design systems, Vitest, Node.js, Electron, Angular

AI: agentic coding with Claude Code, LLM-based tooling for code generation, AI-ready engineering standards for coding agents

Accessibility: WCAG 2.2 AA, European Accessibility Act, keyboard and spatial navigation, screen readers, voice-control support, DOM overlays for Canvas content, accessible-by-design practices

DSP & audio: Digital signal processing, physically-informed sound synthesis and analysis, audio and multimedia computing

Backend & other: Python, Flask, SQLAlchemy, C/C++

Engineering practice: Architecture and technical strategy, large-scale monorepo migrations, ADRs, RFCs, technical documentation, mentoring, technical interviewing

EDUCATION

MSc in Economics · University Niccolò Cusano — 2017 – 2020

Main subjects: Environmental Economics, Statistics, Accounting. Final grade: 110/110.

BSc & MSc in Computer Science for New Media Communications · University of Milan — 2005 – 2011

Main subjects: Software Engineering, Digital Signal Processing, Distributed Systems, Human-Computer Interaction, Music Information Retrieval, Semiotics. Final grade: 110/110 cum laude.

RELEVANT PROJECTS

- **Audio plugins** - independent line of audio plugins built on physically and statistically grounded DSP (parametric and analytical methods), distributed via Polar.sh. https://stefanobaldan.com/plugins/
- **vodit** - minimalist dependency injector. No frills, no dependencies, no decorators. https://singintime.github.io/vodit
- **SDT (Sound Design Toolkit)** - open-source C library for physically-based sound synthesis; developer and maintainer, 2014–2017. https://github.com/SkATVG/SDT
- **MaIs** - performance presented in Milan in 2013. Co-author. https://vimeo.com/86330894
- **Sonic Tennis** - audio-only game for iOS devices. https://youtu.be/0uxqhCS0GPg
- **GlobeMX** - interactive HTML5 media player. Master's thesis project. http://emipiu.di.unimi.it
- **IPCapture** - Processing library for acquiring MJPEG live streams. https://github.com/singintime/ipcapture
- **Musica Sull'Acqua** - sonification of the civic aquarium of Milan. https://ludovico.lim.di.unimi.it/download/papers/SMC2012.pdf
- **Moonlanding** - interactive installation, Festival della Creatività, Florence, 2009. Sound designer. https://youtu.be/kEERdTvCbYI

LANGUAGES

Italian (native) · English (C2) · French (C1) · German (B1)

OTHER

Playing piano and mandolin, singing, recording and mixing in a home studio, folk dancing, juggling and fencing.